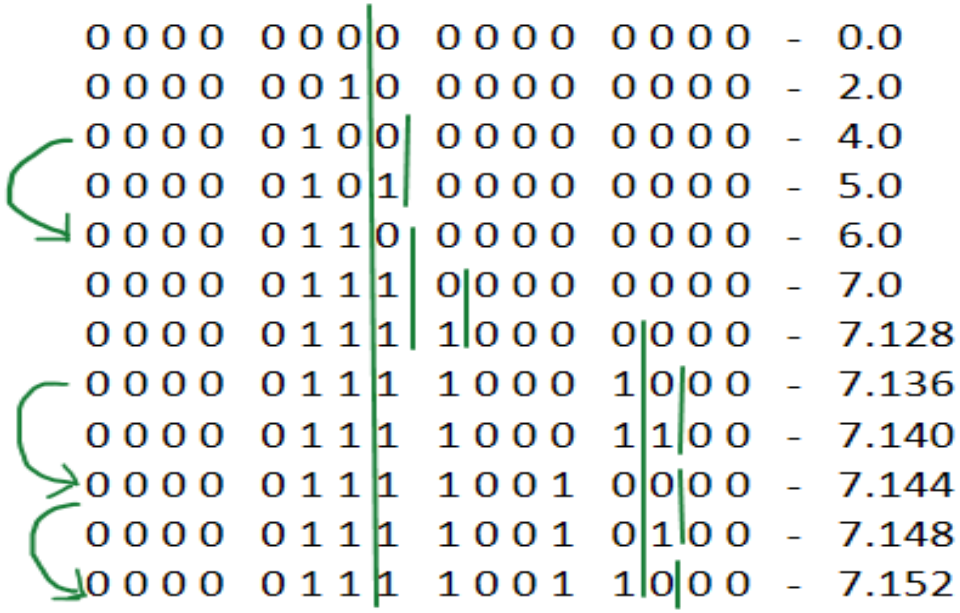
Group: 2703
Topic 8 -> Smart Bangladesh Digital City Network

VLSM Calculation:

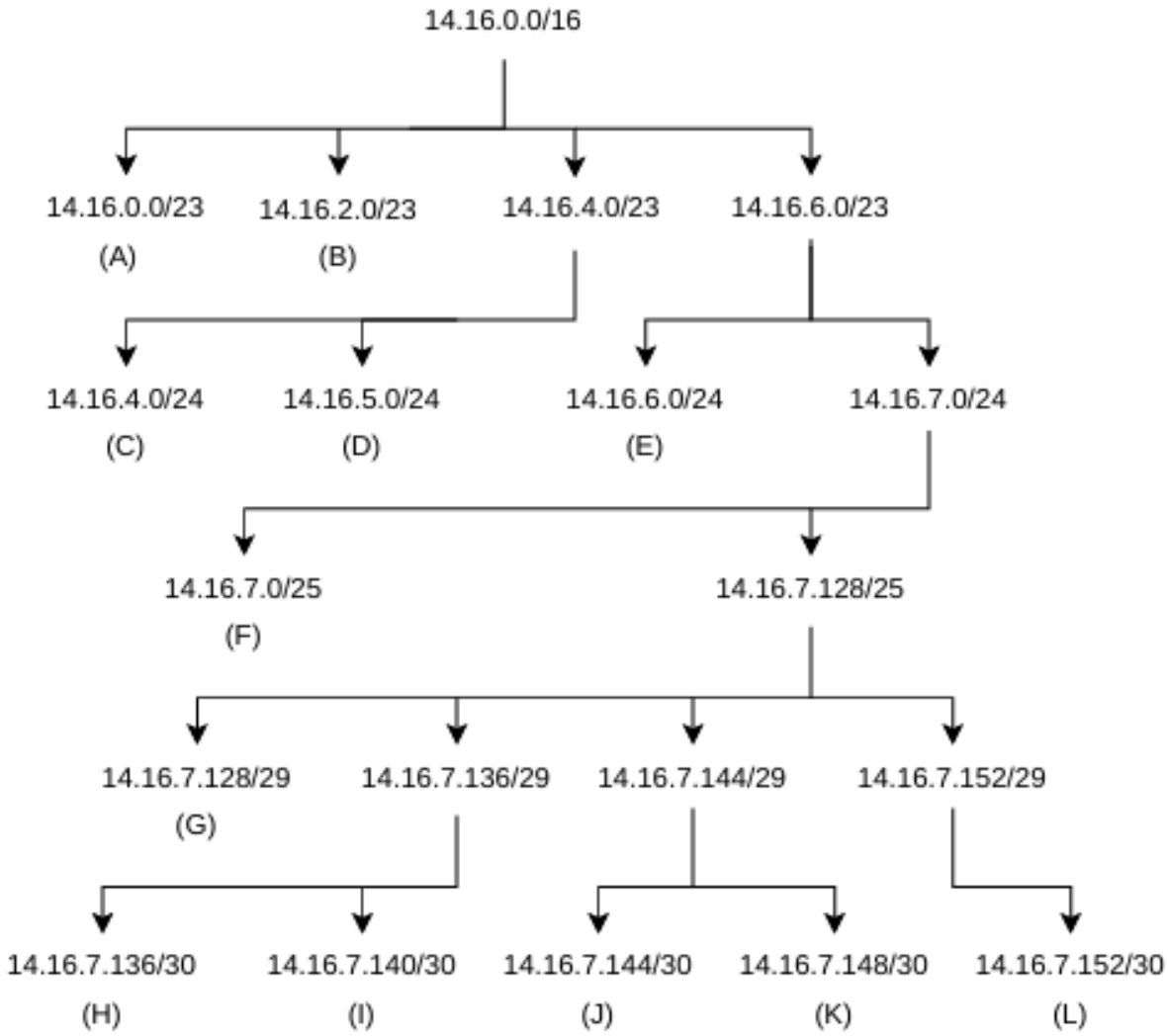


IP Address Table:

Base Network = 14.16.0.0/16

	Department	Host	Host +2	Blocksize	Host bits	Prefix	IP Address
A	SCCC LAN	320	322	512	9	23	14.16.0.0/23
B	DCS LAN	260	262	512	9	23	14.16.2.0/23
C	PHD LAN	220	222	256	8	24	14.16.4.0/24
D	TMA LAN	180	182	256	8	24	14.16.5.0/24
E	ESC LAN	140	142	256	8	24	14.16.6.0/24
F	ERU LAN	100	102	128	7	25	14.16.7.0/25
G	Central switch(SCCC,TMA ,PHD)	3	5	8	3	29	14.16.7.128/29
H	WAN : SCCC-ERU (serial)	2	4	4	2	30	14.16.7.136/30
I	WAN:ERU-DCS	2	4	4	2	30	14.16.7.140/30
J	WAN:DCS-TMA	2	4	4	2	30	14.16.7.144/30
K	WAN:PHD-TMA	2	4	4	2	30	14.16.7.148/30
L	WAN:DCS-ESC	2	4	4	2	30	14.16.7.152/30

VLSM Tree:



Commands Used:

DHCP Configuration:

Router-based DHCP (SCCC-Router serves SCCC, TMA, PHD pools):

ip dhcp excluded-address 14.16.0.1 14.16.0.15
ip dhcp excluded-address 14.16.5.1 14.16.5.15
ip dhcp excluded-address 14.16.4.1 14.16.4.15

ip dhcp pool SCCC_POOL
network 14.16.0.0 255.255.254.0
default-router 14.16.0.1
dns-server 14.16.0.2

ip dhcp pool TMA_POOL
network 14.16.5.0 255.255.255.0
default-router 14.16.5.1
dns-server 14.16.0.2

ip dhcp pool PHD_POOL
network 14.16.4.0 255.255.255.0
default-router 14.16.4.1
dns-server 14.16.0.2

DHCP Relay (ip helper-address, on TMA, PHD, and ESC routers):

! TMA-Router
interface GigabitEthernet0/0
ip helper-address 14.16.7.129

! PHD-Router
interface GigabitEthernet0/0
ip helper-address 14.16.7.129

! ESC-Router
interface GigabitEthernet0/0
ip helper-address 14.16.2.2

DNS Configuration:

On SCCC Server (14.16.0.2) → Services → DNS → ON, with A-records:
www.smartcity.gov.bd -> 14.16.0.3
www.emergency.gov.bd -> 14.16.7.3
mail.tma.gov.bd -> 14.16.5.2
mail.esc.gov.bd -> 14.16.6.2

Email Server Configuration:

TMA Server (14.16.5.2):
domain `mail.tma.gov.bd`,
users `traffic1`, `admin`

ESC Server (14.16.6.2):
domain `mail.esc.gov.bd`,
users `student`, `teacher`

Static Routing:

! SCCC-Router — Hybrid static (feeds backup to TMA/PHD via ERU)
ip route 14.16.7.0 255.255.255.128 14.16.7.138
ip route 14.16.2.0 255.255.254.0 14.16.7.138
ip route 14.16.6.0 255.255.255.0 14.16.7.138
ip route 14.16.5.0 255.255.255.0 14.16.7.138 130
ip route 14.16.4.0 255.255.255.0 14.16.7.138 130

```
! TMA-Router — Floating static (AD 130) via DCS
ip route 14.16.0.0 255.255.254.0 14.16.7.145 130
ip route 0.0.0.0 0.0.0.0 14.16.7.145 130

! ERU-Router — Next-hop static routes
ip route 14.16.0.0 255.255.254.0 14.16.7.137
ip route 14.16.2.0 255.255.254.0 14.16.7.142
ip route 14.16.6.0 255.255.255.0 14.16.7.142
ip route 14.16.5.0 255.255.255.0 14.16.7.142
ip route 14.16.4.0 255.255.255.0 14.16.7.142
ip route 14.16.7.144 255.255.255.252 14.16.7.142
ip route 14.16.7.152 255.255.255.252 14.16.7.142

! DCS-Router — Directly connected static routes (exit-interface only)
ip route 14.16.6.0 255.255.255.0 Serial0/1/0
ip route 14.16.7.0 255.255.255.128 Serial0/0/0
ip route 14.16.0.0 255.255.254.0 Serial0/0/0
ip route 14.16.5.0 255.255.255.0 Serial0/0/1
ip route 14.16.4.0 255.255.255.0 Serial0/0/1

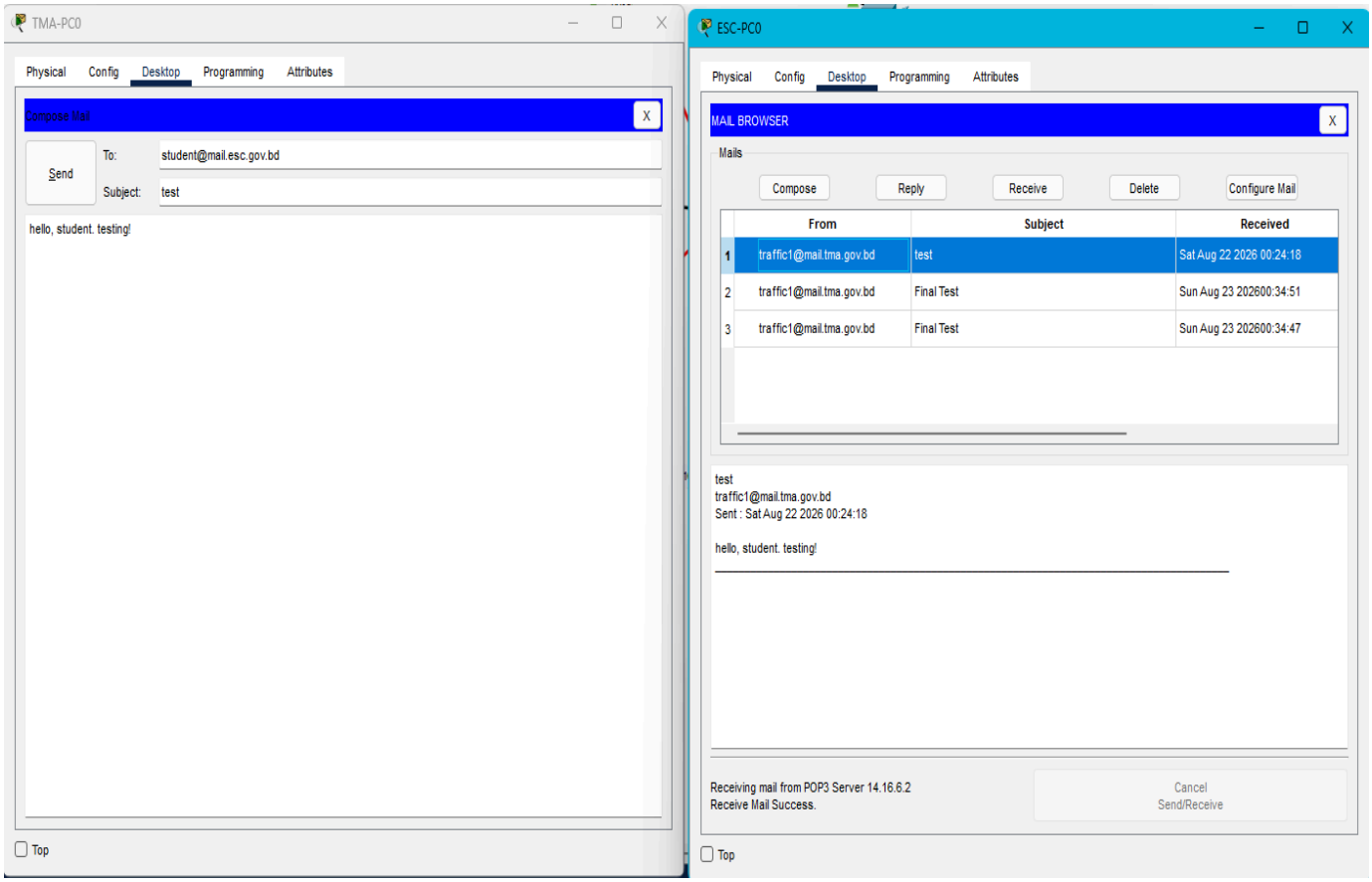
! ESC-Router — Default static route toward DCS
ip route 0.0.0.0 0.0.0.0 14.16.7.153
```

Dynamic Routing (RIPv2):

```
router rip
version 2
no auto-summary
network 14.0.0.0
```

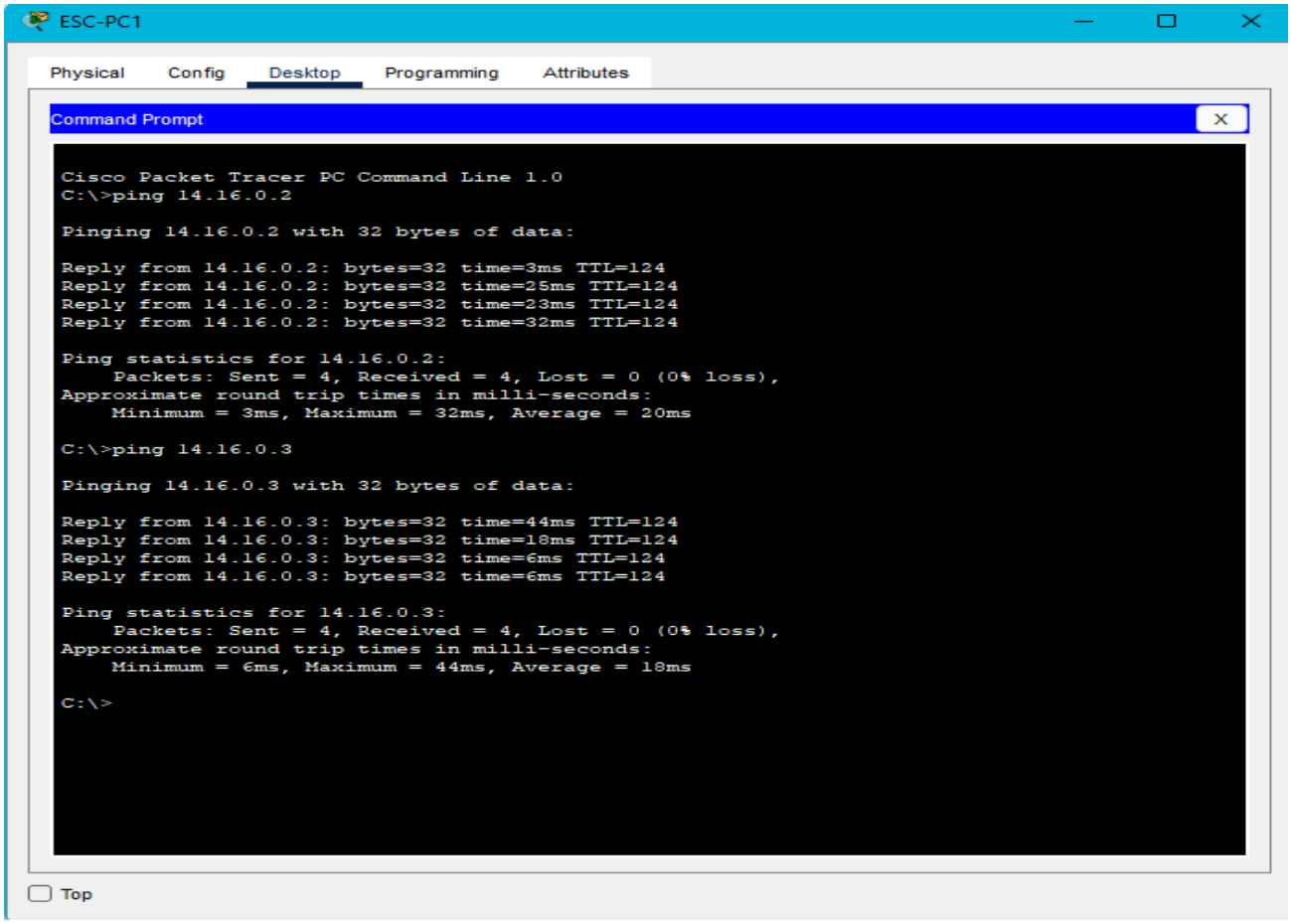
Screenshots for Deliverables:

a) Cross-domain email communication

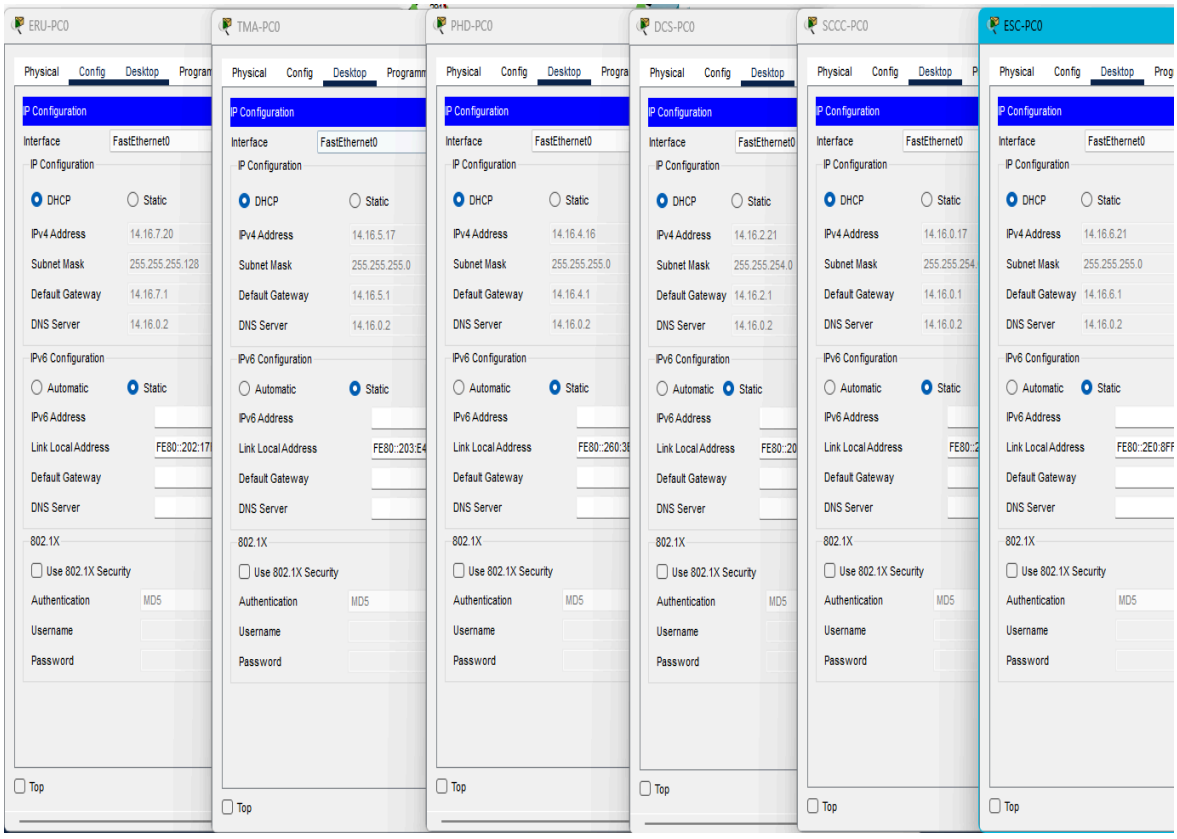


b) Website access using DNS

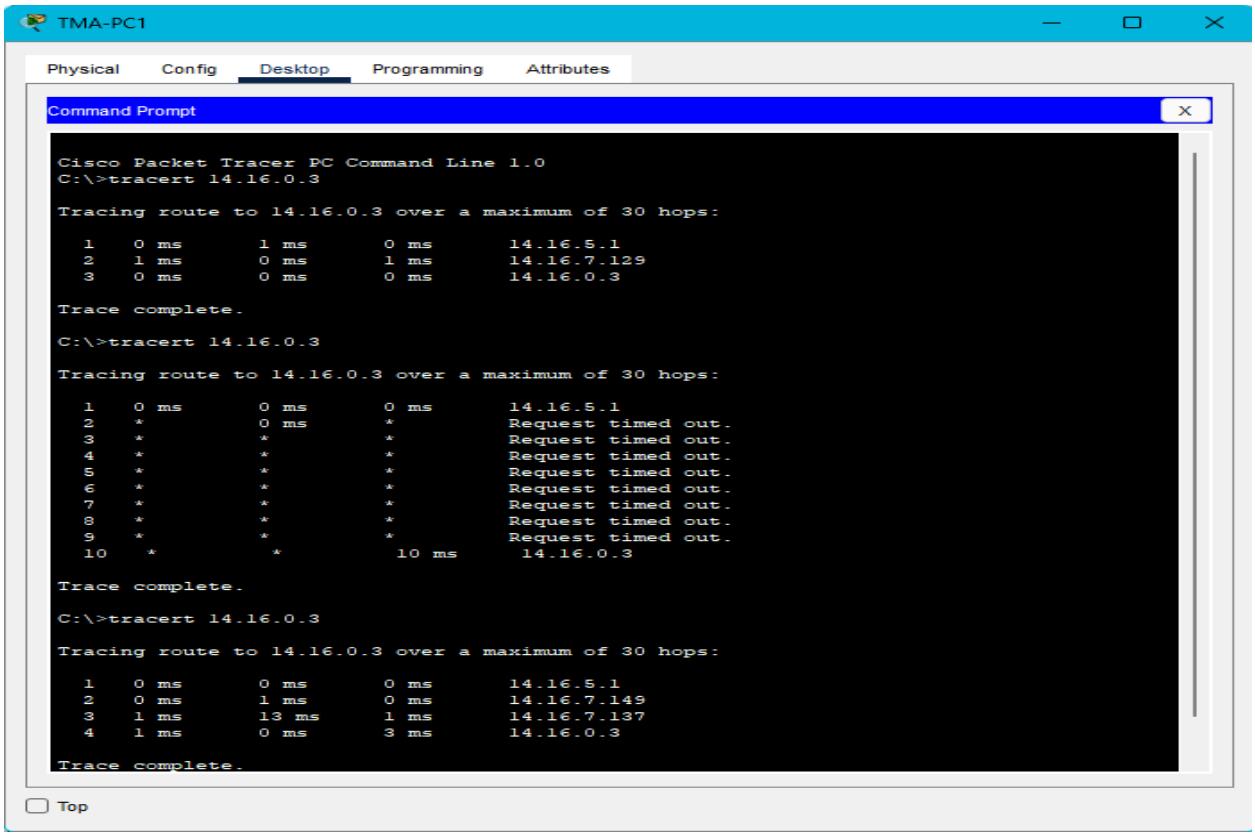




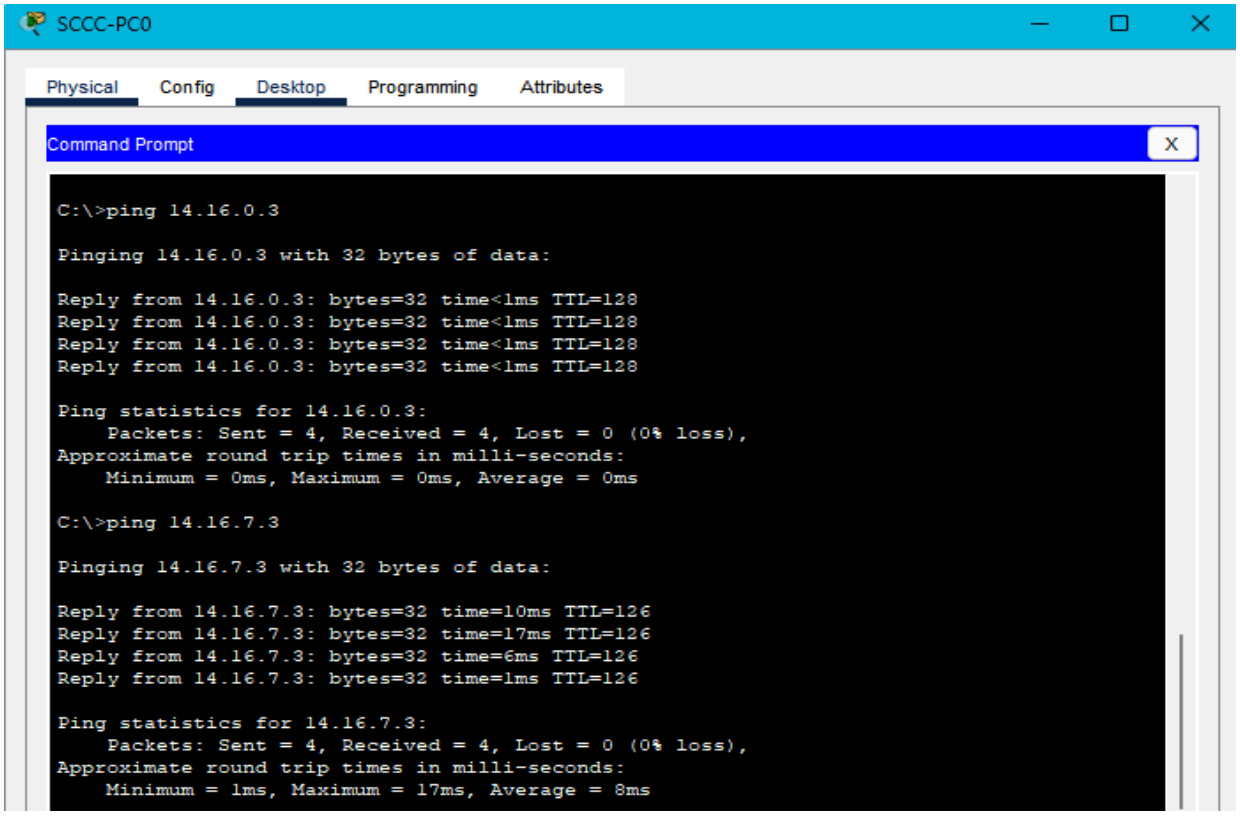
c) DHCP address assignment

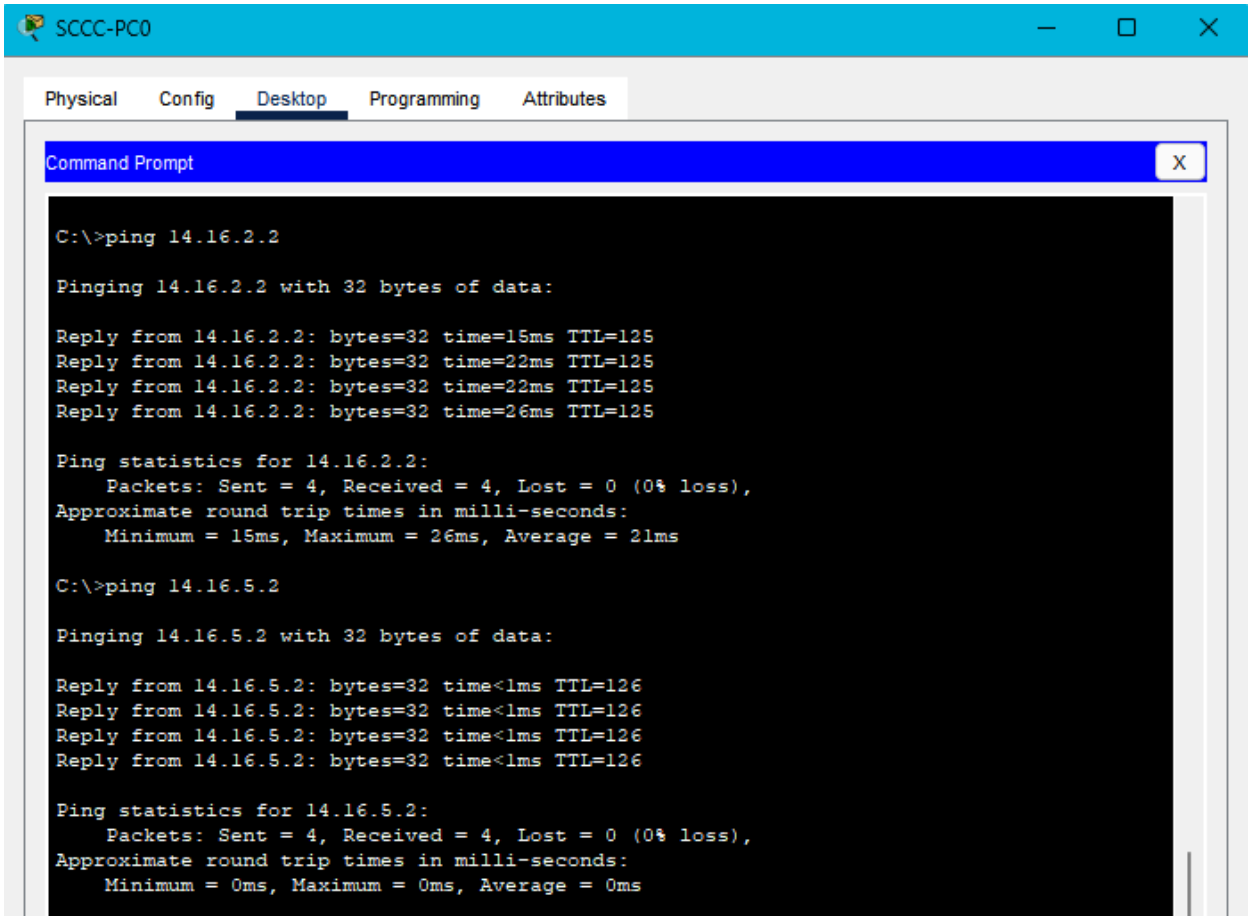


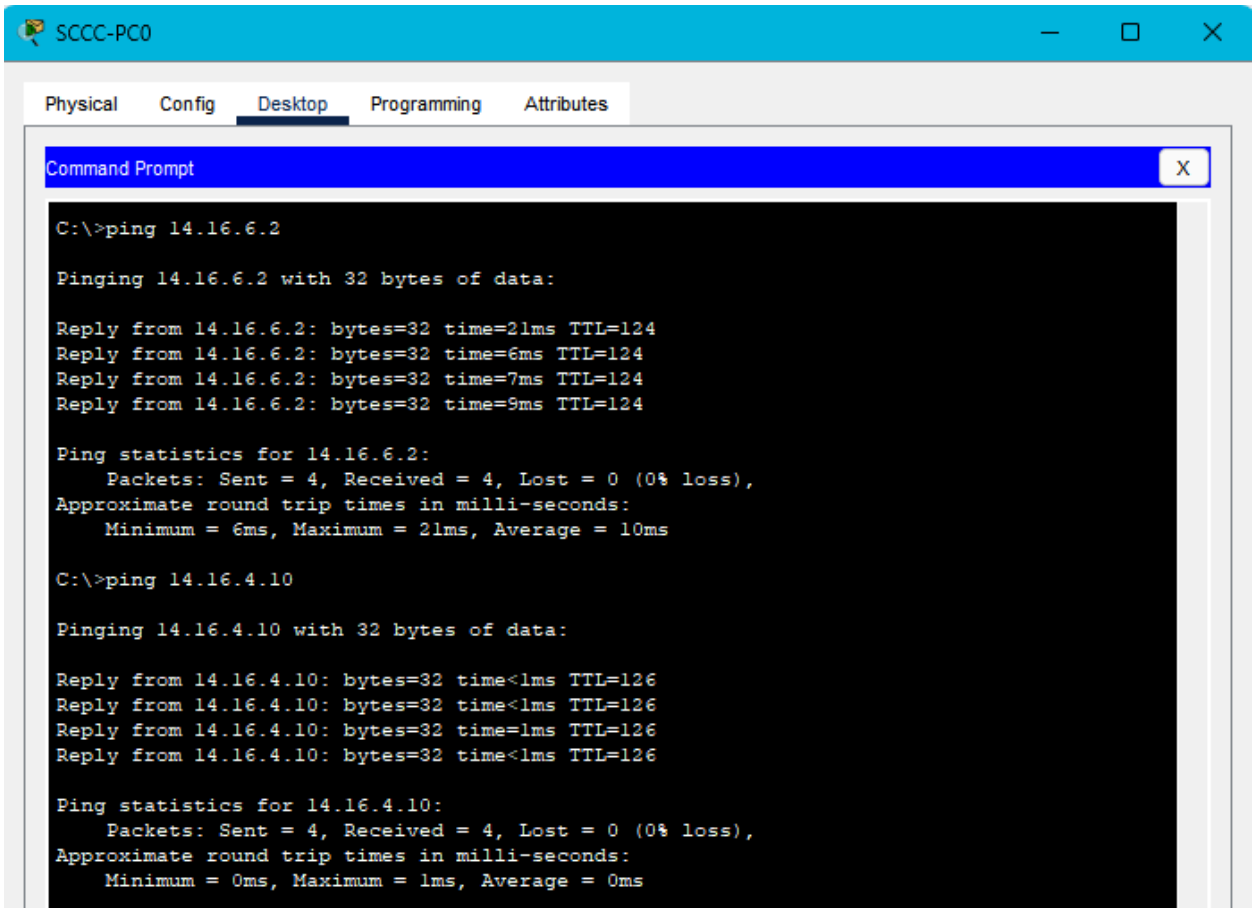
d) Backup route after primary link failure



e) Successful ping tests between all departments




SCCC-PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
C:\>ping 14.16.2.2
Pinging 14.16.2.2 with 32 bytes of data:
Reply from 14.16.2.2: bytes=32 time=15ms TTL=125
Reply from 14.16.2.2: bytes=32 time=22ms TTL=125
Reply from 14.16.2.2: bytes=32 time=22ms TTL=125
Reply from 14.16.2.2: bytes=32 time=26ms TTL=125
Ping statistics for 14.16.2.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 15ms, Maximum = 26ms, Average = 21ms
C:\>ping 14.16.5.2
Pinging 14.16.5.2 with 32 bytes of data:
Reply from 14.16.5.2: bytes=32 time<1ms TTL=126
Reply from 14.16.5.2: bytes=32 time<1ms TTL=126
Reply from 14.16.5.2: bytes=32 time<1ms TTL=126
Reply from 14.16.5.2: bytes=32 time<1ms TTL=126
Ping statistics for 14.16.5.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms


SCCC-PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
C:\>ping 14.16.6.2
Pinging 14.16.6.2 with 32 bytes of data:
Reply from 14.16.6.2: bytes=32 time=21ms TTL=124
Reply from 14.16.6.2: bytes=32 time=6ms TTL=124
Reply from 14.16.6.2: bytes=32 time=7ms TTL=124
Reply from 14.16.6.2: bytes=32 time=9ms TTL=124
Ping statistics for 14.16.6.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 6ms, Maximum = 21ms, Average = 10ms
C:\>ping 14.16.4.10
Pinging 14.16.4.10 with 32 bytes of data:
Reply from 14.16.4.10: bytes=32 time<1ms TTL=126
Reply from 14.16.4.10: bytes=32 time<1ms TTL=126
Reply from 14.16.4.10: bytes=32 time=1ms TTL=126
Reply from 14.16.4.10: bytes=32 time<1ms TTL=126
Ping statistics for 14.16.4.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms